

Synthetic Forecasting Walkthrough

Question and evidence

How does a simple trend compare with a last-value baseline? This is a deliberately constructed software fixture, not a competition paper or a model effectiveness study. All values are synthetic and saved in result.json.

Model and split

Only observations 1-4 fit the trend. Observations 5-6 are held out. The average step is computed from the first and last training values. The baseline repeats the final training value.

$$\hat{y}_{t+h} = y_t + h \frac{y_t - y_1}{t - 1}$$

Observation	Split	Actual	Trend forecast	Last-value baseline
1-4	Training	2, 4, 6, 8	Not evaluated	Not evaluated
5	Holdout	10	10	8
6	Holdout	12	12	8

Computed result

Trend MAE: 0.0. Baseline MAE: 3.0. Each value is bound to a JSON pointer and source hash. The trend matches because the fixture was constructed to be linear; this is not evidence of generalization.

What the workflow checks

The authoring audit checks the evidence hash, claim values, section coverage and model-review record. Build produces LaTeX; render compiles a PDF. A separate visual review is still required before verify can pass.

What remains outside the example

Real data provenance, alternative models, uncertainty and independent mathematical review are not demonstrated. Do not reuse these numbers in a research conclusion. Changing result.json requires regenerating its evidence hash and rebuilding downstream outputs.